

Code-Layout, Readability and Reusability

Field	Value
Date	05 July 2026
Team ID	(Your Team ID)
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in all source files.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for dataset, models, backend, frontend, and utilities.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named based on their functionality.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections are documented with comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Individual modules developed for prediction and recommendation tasks.

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7	No Redundant Code	Duplicate or unused code is removed	Yes	Common functions are reused across modules.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation and exception handling implemented.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python	Crop & Yield Prediction	High
2	Crop Recommendation Model	Python (Machine Learning)	Recommendation Engine	High
3	Fertilizer Recommendation Module	Python	Recommendation System	High
4	Weather Data Processing	Python	Prediction Modules	Medium
5	Dashboard UI Components	HTML, CSS, JavaScript	User Interface	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules.
Readability	5	Meaningful names and proper formatting improve readability.

Aspect	Rating (1-5)	Comments
Reusability	5	Reusable modules minimize duplicate code.
Documentation / Comments	4	Adequate comments added for major functions.
Overall Score	4.75 / 5	Code is modular, readable, maintainable, and reusable.